DREAM - Danish Research Institute for Economic Analysis and Modelling

GreenREFORM EU Version 2.0

PART A – BACKGROUND AND INTRODUCTION

1. Who will implement the project?

Briefly describe the main implementing organisation(s) and its/their mission. If this project is a proposal for continued support from KR Foundation, please provide a brief update on the work supported by any ongoing or previous KR Foundation grants.

DREAM specializes in research-based analyses of the Danish economy, covering core areas such as macroeconomics, fiscal sustainability, demographics, education, climate and energy, the housing market, the labour market, and pensions. Our analyses serve a wide range of stakeholders, including ministries, commissions, NGOs, and think tanks. As a non-profit organisation, DREAM is primarily funded through a financial contribution from the Danish Ministry of Finance. Additionally, revenue generated by research and analyses conducted for various clients contributes to our funding.

In recent years, DREAM has undertaken three international projects. Most recently, DREAM has been developing a new climate-economic model for use in four EU member states (see Section 2). This project builds on experience with the GreenREFORM model, a climate and environmental economic dynamic computable general equilibrium (CGE) model[1] of the Danish economy developed by DREAM in collaboration with researchers from the University of Copenhagen, Aarhus University and the Technical University of Denmark. The development of GreenREFORM was supported by the KR Foundation as part of the foundation's grant to the Green GDP project initiated by researchers from the University of Copenhagen. The GreenREFORM model was recently used by the Danish government's Expert Group on Green Tax Reform and is currently applied by several Danish ministries and think tanks.

2. What are the key issues addressed?

Describe the key issue(s) that will be addressed. Note: There is no need to explain the climate crisis and why it is a challenge; please focus specifically on the key issue(s) that the project will address, e.g., expansion of fossil fuel infrastructure, misinformation, greenwashing, etc.

Climate policy is broadly recognized as a core issue by Ministries of Finance (MoF), but very few have the analytical capacity to perform integrated analyses of economic and environmental policy. This is evident in a recent survey conducted by The Coalition of Finance Ministers for Climate Action.

[1] GreenREFORM is a dynamic computable general equilibrium (CGE) model designed to integrate economic and environmental policy analyses in Denmark. The combination of advanced features in GreenREFORM is unique: Forward looking behavior, credible short run dynamics, an integrated welfare economic measure, detailed energy and emissions accounts, and a comprehensive bottom-up representation of technological emission abatement. This enables better analyses of the transitional impacts of climate policy and of the interactions between structural and technical changes, than other currently available analytical tools. For this reason, GreenREFORM has attracted a lot of international attention.

If MoF's are not able to analyse the economic impacts of climate change or the risks and opportunities of decarbonisation, they cannot plan or budget for it, let alone help shape an effective policy response. Economic policy thus risks remaining a key obstacle to climate policy.

The proposed project addresses this issue head on. GreenREFORM EU is a simplified version of the original GreenREFORM model for Denmark, prepared for implementation in other countries and adaptable to their particular needs. As has been the case in Denmark (see Section 3), the GreenREFORM EU model will support the design of climate policy in the four countries, by providing a solid foundation for informed decisions on climate policy.

The first version of the the GreenREFORM EU model (Version 1.0) is already in development through an ongoing project. That project is sponsored by the European Commission under the Technical Support Instrument (TSI) programme 2024, hence the name GreenREFORM EU. The model development is progressing well, with four out of the original five participating institutions actively working on their own datasets. These include the Ministries of Finance of Austria, Finland, and Italy, as well as the Federal Planning Bureau of Belgium.

PART B: PROJECT SCOPE AND OBJECTIVES

3. What is the project's vision for success?

Provide a brief overview of the project's vision and the long-term impact that the project seeks to achieve.

Our vision is to enhance the capacity for climate policy analysis within the participating institutions by using a unified framework, while making the model available as open source to encourage broader engagement. In the long term, we aim to foster cooperation and knowledge sharing among institutions across countries, ultimately improving the design of national climate policies worldwide.

The recent experience in Denmark, where the GreenREFORM model was utilized to facilitate public debate and political negotiations, culminating in a significant reform of agricultural emissions, exemplifies how economic policy can be transformed from an obstacle into an enabling factor for climate policy. Using GreenREFORM, the Expert Group for a Green Tax Reform showed how legitimate concerns about public finances, local jobs, land price depreciation, economic growth, etc. could be addressed in a balanced manner, paving the way for broad political support for reform[2].

4. What is the project's theory of change?

Illustrate the project's theory of change and explain why it is believed to be effective. Note that elements in the theory of change (e.g., objectives, activities, and outputs) are to be unfolded in the following questions.

We believe in leading by example through successful small-scale projects. In that spirit, this project aims to build on the momentum gained in the ongoing TSI-sponsored project, and help prevent a stop-and-go process, which could ultimately lead to the abandonment of the project in one or more of the participating institutions, and possibly to the larger vision of bringing GreenREFORM out into the world.

[2] GreenREFORM was used extensively by the Expert Group for a Green Tax Reform and later by the Government itself to inform the public debate and political negotiations leading to the Green Triparte Agreement of 2024.

While the interest in the Danish experience and the GreenREFORM model in particular is global and widespread, timing and momentum are critical factors for success. The state of development of the GreenREFORM model and the momentum of climate policy kickstarted by the 2019 elections, which ultimately led to where we are today, is at least to some extent a happy coincidence. Likewise, in the case of the ongoing development of GreenREFORM EU Version 1.0, the timing of our ability to deliver on the present demand of the four participating institutions was critical.

We thus believe strongly in the importance of building on the current momentum of the TSI-sponsored project. If we keep pushing model development and continue providing practical support to the four institutions, we can pave the way for better and more effective climate policy in the four participating countries and beyond, just as we have done in Denmark.

The 'GreenREFORM EU Version 2.0' project will not only help promote climate policy in the four participating countries, but will also serve as a crucial stepping stone toward future projects with greater outreach. To promote future engagement, especially from lower-income countries, we will make the GreenREFORM EU Version 2.0 model available as open source.

Looking beyond the horizon of this project application, the four participating institutions and DREAM are already working on preparing an application for a more ambitious project for the 2026 TSI programme, aimed at developing country-specific submodels for GreenREFORM EU. If both the current project application and the TSI 2026 application are successful, this will ensure a smooth transition with continuous model development and support for the capacity-building efforts of the participants. Simultaneously, at least one other project aimed at the implementation of GreenREFORM EU in a developing country is also taking shape and looking promising.

5. What are the project's key objectives?

List the project's key objectives in bullets, including the associated indicators of success. The indicators must be SMART (Specific, Measurable, Achievable, Realistic and Timely).

Objective: Enhance the capacity for climate policy analysis within the four participating institutions.

- Indicator: The Federal Planning Bureau of Belgium will have completed or be engaged in analyses based on the GreenREFORM EU model by the project's end.
- Indicators: The three other institutions will maintain momentum, ensuring that by the project's conclusion, they each have a model calibrated to their own data, and within 1-2 years, they too will have completed analyses based on the model.

Objective: Foster engagement from additional countries and support the design of new, more ambitious projects.

- Indicator: A refined and well-documented GreenREFORM EU Version 2.0 will be publicly available.
- Indicator: By the end of the project, at least one project with new partner countries – preferably developing countries – will be initiated or be in the final stages of initiation.

6. What activities are planned to achieve the objectives?

Describe the specific activities and outputs proposed to realise the key objectives, including a clear plan and timeline for implementation. If communication is a key activity, describe the strategy and plan for the communication activities, including target groups and choice of communication channels. Note: Do not just list the activities in bullets but explain them in the context of the theory of change.

In the spirit of our theory of change (Section 4), the 2024 TSI project is designed with a narrow scope and a short project period of just one year. While the project is progressing as expected, continued support for the four institutions and further model development are thus essential for the project's ultimate success. The ongoing project concludes in December 2025, and the current project proposal seeks to extend the project by one year.

Project activities

The activities are strategically designed to operationalize our theory of change, which emphasizes leading by example through successful small-scale initiatives. By refining and promoting GreenREFORM EU, we aim to empower the four participating institutions to enhance their capacities for climate policy analysis. This includes creating user-friendly features and comprehensive documentation, ensuring accessibility and usability.

Capacity-building workshops and individualized support will foster ongoing engagement and skill development, equipping institutions with the necessary tools to implement the model effectively.

Additionally, by making the refined model open source, we encourage participation from additional countries, particularly lower-income regions. This approach aligns with our objective of promoting future projects and partnerships. The iterative feedback mechanisms incorporated into our activities will ensure continuous improvement and adaptation, ultimately leading to a robust framework for climate policy analysis that can be scaled and replicated in diverse contexts.

In summary, our activities are interconnected and directly support our long-term goals of capacity enhancement, stakeholder engagement, and sustainable development in climate policy.

Model development:

- Refine the model with a focus on user-friendliness and accessibility.
- Engage in iterative feedback with participating institutions.
- Document the theoretical framework of the model.
- Create a practical user guide for the model.
- Publish Version 2.0 of the model online and promote it to potential stakeholders.

Capacity Building.

- Conduct online workshops and training sessions for the four participating institutions.
- Provide individualized support and guidance to each institution as needed.
- Organize a series of online presentations for new personnel in the participating institutions and potential stakeholders in future projects to introduce the model and encourage future project development.

Data and implementation

- Engage with each of the four participating institutions individually to offer tailored support and guidance.

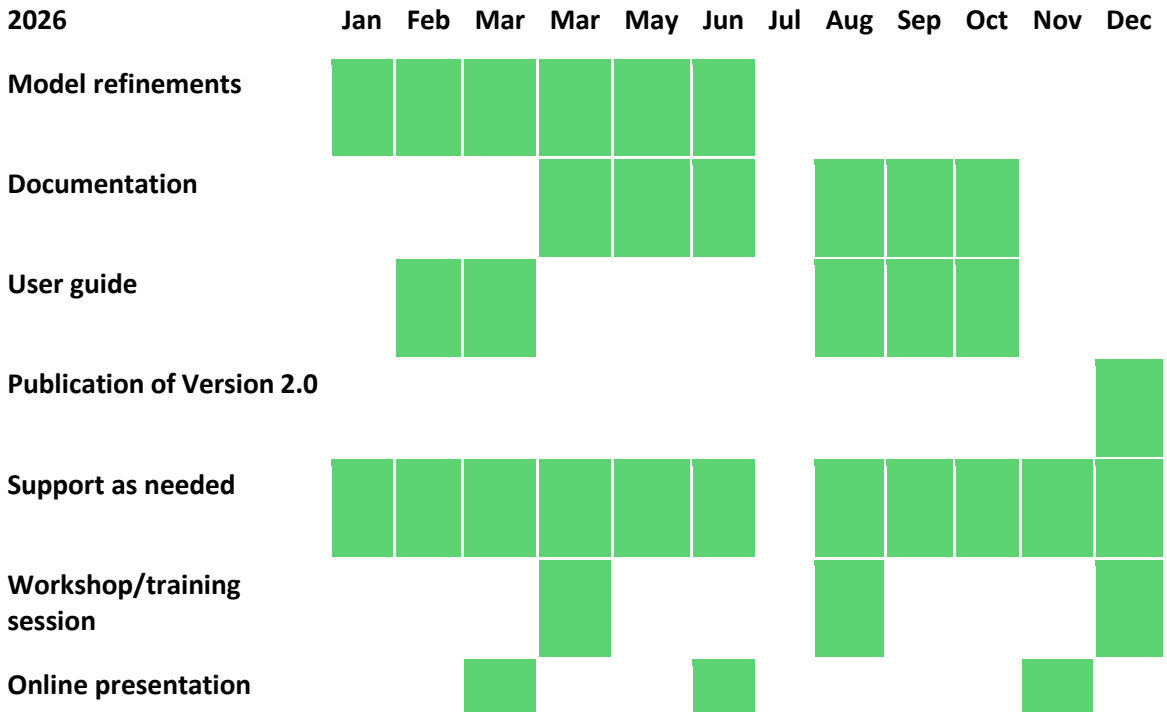
Timeline of activities

The project is planned to start in January 2026 in direct continuation of the ongoing TSI 2024 project, which ends in December 2025. The project period is one year, with a potential small overlap with the (hopefully) forthcoming TSI 2026 project, which will commence in the later part of 2026.

The project plan needs to be flexible at this initial stage, to accommodate the actual needs of each of the four participants, as they move forward at different paces and with varying priorities.

At the initial phase of the project, each of the four institutions will be asked to produce an updated plan for data development and implementation, identifying the need for support and enabling a revision of the project plan.

A timeline of activities is presented below. The expectation is that the first half of the project period will be primarily focused on model development and direct support activities, while in the second half of the project period, attention will be directed towards producing the material aimed at publication by the end of the project, i.e. a refined version of the model and the associated theoretical documentation. Training/workshop sessions and introductory online presentations will be planned, expectedly about three months apart throughout the project period.



7. What are the key organisational competencies, and who are the key partners?

Describe how the organisation, based on its organisational capacity as well as partnerships and collaborations with other organisations, is well positioned to implement the project. If the organisation has a proven track record with this type of work, briefly outline any relevant key achievements. Relevant expertise of key experts related to the project should be included in the supporting documentation B) "biographies of key personnel".

DREAM possesses extensive experience in building macroeconomic models, particularly Computable General Equilibrium (CGE) models. Our team consists of about 30 highly skilled model developers working collaboratively in one office. The expertise gained from our close collaboration with government civil servants and university researchers in the development and implementation of analytical tools is unique. In addition to the TSI project, DREAM has recently developed two models for the Danish Ministry of Finance—MAKRO and GreenREFORM—as well as model systems for the Greenland Ministry of Finance and the Fiscal Council of Slovenia.

Statistics Denmark is a key partner in the TSI project, having developed the benchmark data for Denmark and actively supporting the data development for the participating institutions. We anticipate inviting Statistics Denmark to participate in the current project as a subcontractor.

For the past four years, DREAM and the Danish MoF have engaged in close collaboration through the aforementioned Coalition of Finance Ministers for Climate Action and the related Green Macroeconomic Modelling Initiative. In the coalition, the Danish MoF is co-leading the Helsinki principle 4 workstream, which aims to help MoF's and the institutions that support them build capacities to deliver sound economic analysis for effective decision-making. The development and practical implementation of GreenREFORM in the Danish MoF is in many ways the prime example of excellence in this regard. The two forums will be used to showcase the project, but the project should also be seen as a direct means to support the Helsinki principle 4 workstream: The project will offer a state-of-the-art, yet practically accessible, toolkit as open source, and it will showcase how institutions can get started.

PART C: RISKS AND ENVIRONMENTAL IMPACT

8. What are the risks and how will they be managed?

Describe any known risks to achieving the project's objectives and how these risks will be addressed.

The primary risks include staff turnover and changes in priorities among the four participating countries. The project relies on the specialized skills of model developers who were involved in the TSI project. Our aim of avoiding a stop-and-go process is designed to maintain and enhance internal competencies within our own as well as the participating institutions. Nonetheless, DREAM is well equipped to manage staff turnover due to our size and expertise. Changes in priorities among participating institutions are more challenging. For instance, during the TSI project, one research institution withdrew due to a reallocation of resources. However, the remaining institutions—particularly the Federal Planning Bureau of Belgium—appear to remain committed.

9. How will you minimise the environmental impact of the project?

Outline how any associated environmental impacts will be minimised, particularly emissions relating to travel and meetings.

All project activities are planned to occur either in local offices or online. Since funds are allocated primarily for salaries rather than for equipment purchases, the overall environmental impact will be minimal.